



Agent-Based Model v2.0 for Food Security Analysis

Simulating consumer behaviour, supply chains & policy in Finnish grocery retail

Horizon Europe SecureFood | Grant Agreement No. 101136583

What's inside

1	Data & Setup	Load DCE cohort, explore archetypes & product catalogue
2	Simulation	Interactive Demo + Monte Carlo Scientific Analysis
3	Analysis	Food Waste / Per-Product / Behavioural Theory / Agent Replay / Compare
4	Policy & Strategy	Interventions / Stress Test / Multi-Store Network / Regional Map
5	Export	CSV downloads, full data bundle & branded 7-section PDF report
*	SecureFood Simulator	Climate-disruption dairy scenario for Horizon Europe partners

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1. Getting Started - Navigation

GROCERYsim ABM v2.0 uses a 5-card visual navigation system. When you open the app you see the Home screen with five large coloured cards. Each card expands into one or more sub-sections. A clickable breadcrumb at the top of every sub-section lets you jump back to the Home screen instantly.

1.1 The Home Screen - 5 Cards

Card 1 Data & Setup	Load your DCE participant cohort, view archetypes and product catalogue.
Card 2 Simulation	Run the ABM: Interactive Demo (live animation) or Monte Carlo batch.
Card 3 Analysis	Deep-dive results across 6 specialised sub-sections.
Card 4 Policy & Strategy	Run interventions, stress tests, multi-store networks, and the regional map.
Card 5 Export	Download CSVs, the full data bundle, and the branded PDF report.

TIP Click any sub-section button inside a card to open it. The breadcrumb at the top (e.g. 'Menu > Analysis > Agent Replay') is fully clickable - click 'Menu' to return to the Home screen, or click the card name to see a description of that card.

1.2 SecureFood Scenario Simulator

The 'Scenario Simulator' button at the very top of the page (next to the GROCERYsim title) opens the dedicated SecureFood climate-disruption tool for Horizon Europe partners. It runs independently of the main 5-card navigation. A 'Scenario Instructions' button next to it downloads a PDF walkthrough.

1.3 Onboarding Tour

Click 'Tour' in the left sidebar at any time to replay the 8-step guided tour. Each step explains a card or feature. Use the Previous / Next buttons to move between steps, or 'Skip tour' to dismiss.

2. Sidebar Parameters Reference

All simulation levers are in the left sidebar. Changes take effect on the next run.

2.1 Core Simulation

Simulation Days	Number of days to simulate (default 60). Longer runs reveal recovery dynamics.
Number of Agents	Consumer agents per day. Higher counts improve statistical stability but increase run time.
Store Size	S / M / L - auto-scales shelf capacity, reorder point, and initial stock levels.
Random Seed	Fix the seed for reproducible results; leave blank for a random seed each run.

2.2 Supply Chain

Reorder Point	Stock level (units) that triggers a replenishment order.
Lead Time (days)	Days between order placement and shelf arrival.
Order Quantity	Units ordered per replenishment cycle.
Supply Disruption	Days the supply chain is fully disrupted during the crisis window.

2.3 Crisis Settings

Crisis Start / End	Day range for the crisis shock. Baseline runs ignore this range.
Inflation Rate (%)	Price inflation applied to all products during the crisis window.
Panic Sensitivity	How strongly consumers react to low stock levels (0-1 scale).
Hoarding Multiplier	Scales how much extra stock panic-buyers attempt to purchase.

2.4 Behavioural Levers

Media Channel	Primary communication channel: social media, TV, radio, or mixed.
Purchase-Limit Nudge	Enable a per-visit purchase cap on high-demand SKUs.
Cap (units/visit)	Maximum units a consumer can buy per product per visit when nudge is active.
Stockpile Horizon	Days ahead consumers plan when hoarding behaviour kicks in.

TIP The Behavioural Theory sub-section in Card 3 - Analysis visualises how these levers affect Prospect Theory loss aversion, TPB intention scores, and FIES food insecurity.

3. Card 1: Data & Setup

This card is the starting point for every session. It contains one sub-section: Data & Population.

3.1 Loading Data

- Firebase (live): If Firebase credentials are configured, the app auto-loads the latest DCE survey cohort on startup. A spinner confirms the live connection.
- CSV / JSON upload: Use the file uploader to load an exported DCE dataset manually. Required columns: age, income, education, archetype, and product preference weights.
- Demo data: If no data is loaded the app generates a synthetic 116-respondent cohort matching the Finnish DCE study parameters.

3.2 Cohort Overview

- Demographics panel: age distribution, income brackets, education levels.
- Archetype clustering: K-Means (k=4) segments consumers into Bargain Hunter, Quality Seeker, Health Conscious, and Convenience Shopper.
- DCE preference weights: mean willingness-to-pay for Finnish origin, organic, low-fat, and packaging attributes from the survey data.

3.3 Product Catalogue

The bottom of the Data tab displays the full product catalogue: SKU names, categories (Dairy / Meat / Produce / Packaged), shelf life, base price, and current stock levels. The catalogue drives the per-product analysis in Card 3.

TIP Always load or verify data before running a simulation. The demo cohort is used automatically if no file is uploaded, so you can explore the tool without any data file.

4. Card 2: Simulation

Card 2 contains two sub-sections: Interactive Demo and Scientific Analysis.

4.1 Interactive Demo

- Runs a paired Baseline vs Crisis simulation using your sidebar parameters.
- Day-by-day animation: watch revenue, shelf stock, and consumer panic evolve in real time.
- Adjust Animation Speed (seconds per day) in the sidebar for slower or faster playback.
- Metrics panels update each day: daily revenue, units sold, stockout events, waste, lost sales.
- Save scenario: give the current run a name and click 'Save' to store it for later comparison.
- Both Baseline and Crisis models share the same random seed for a fair paired comparison.

TIP Run the Interactive Demo first after changing parameters - it populates the data required by Food Waste, Per-Product, Behavioural Theory, and Agent Replay in Card 3.

4.2 Scientific Analysis (Monte Carlo)

- Runs N independent replications of the paired simulation (default N = 20).
- Displays p10 / IQR (p25-p75) / p90 percentile confidence bands on revenue charts.
- Baseline vs Crisis dual-band chart: directly overlays both scenarios' uncertainty ranges.
- AI storage recommendations: interprets results and gives plain-language advice.
- Statistical comparison table: mean revenue, waste, lost sales for both scenarios.
- PDF Report available from the Export card once a run is complete.

5. Card 3: Analysis

Card 3 has six sub-sections. All require a completed Interactive Demo run. Navigate to each using the card's sub-section buttons on the Home screen.

5.1 Food Waste

- Daily waste log: units expired per product per day for both scenarios.
- Waste drivers bar chart: which SKUs generate the most waste.
- CO2 footprint: converts wasted units to kg CO2-equivalent using product emission factors.
- Baseline vs Crisis waste trajectories on a shared axis.

5.2 Per-Product Analysis

- Select any SKU from the dropdown to view its stock, sales, and price over time.
- Stockout frequency chart: number of days each SKU was at zero stock.
- Price evolution: how crisis inflation affects individual product prices.

5.3 Behavioural Theory

- Prospect Theory: loss aversion coefficient over time - peaks during the crisis window.
- TPB intention score: Theory of Planned Behaviour composite (PBC + Subjective Norm).
- FIES scale: Food Insecurity Experience Scale index by income bracket (FAO 2016).
- Stockpile pressure: aggregate consumer hoarding pressure index by day.
- Use the Behavioural Levers sidebar to see how media channel and nudges shift outcomes.

5.4 Sensitivity Analysis

- Parameter sweep: automatically varies each sidebar lever over a defined range.
- Tornado chart: ranks parameters by their impact on mean revenue, waste, and lost sales.
- Interaction plot: shows how pairs of parameters jointly affect outcomes.

5.5 Compare Scenarios

- Side-by-side charts for any two saved scenarios from the Interactive Demo.
- Revenue, waste, lost sales, and panic level overlaid on shared axes.
- Saved scenarios persist within the session; use Export to download them as CSV.

5.6 Agent Replay Viewer

- Day slider: step through the simulation day by day after running the demo.
- Scatter plot: every shopper's Panic Level (x) vs Perceived Behavioural Control (y). Colour = archetype. The red dashed box marks the high-panic / low-PBC danger zone.
- Archetype breakdown bar chart: daily share of each consumer segment.
- Income vulnerability heatmap: fulfilment rate by income bracket and day.

TIP Run the Interactive Demo, then open Agent Replay. Use the day slider to pinpoint exactly when panic peaks and which archetypes are most affected.

6. Card 4: Policy & Strategy

Card 4 has five sub-sections for intervention analysis, stress testing, and network simulation. Each can be opened independently from the Home screen.

6.1 Policy Analysis

- Fat tax: apply a percentage tax to high-fat products and measure demand shift.
- Domestic / organic subsidy: reduce price for Finnish-origin or organic SKUs.
- Purchase-cap nudge: set a per-visit limit on hoarding-prone products.
- Nutritional labelling: toggle front-of-pack labelling effect on health-conscious consumers.
- Each policy run generates a branded PDF policy brief (downloadable from this sub-section).

6.2 Stakeholder View

- KPI dashboard: large metric cards for revenue change, food security index, waste reduction.
- Policy narrative: plain-language summary suitable for decision-maker briefings.
- Toggle between Supply Chain Actor and Policy Maker perspective.

6.3 Stress Test

- Automated 6-scenario battery: Supply Chain Collapse, Price Spike (+40%), Panic Wave, Import Crisis, Demand Surge, Cold-Chain Failure.
- Each scenario runs a full simulation and records peak revenue loss, max panic level, and recovery time (days to return within 5% of baseline).
- Results table ranks scenarios by severity - sortable by any column.
- Identifies which type of shock your store is most vulnerable to.

6.4 Multi-Store Network Simulation

- Configure up to 8 Finnish stores in an editable table: name, location, size, proximity to other stores, and whether it is a distribution hub.
- Panic contagion: elevated panic in one store decays with distance and spreads to neighbours.
- Emergency redistribution: hub stores can transfer surplus stock to depleted stores.
- Daily revenue by store is shown in a vertical legend chart (avoids overlap for up to 8 stores).
- Per-store summary table: risk level, revenue loss %, and peak panic for each store.
- Click 'View Results on Regional Map' to overlay risk levels on the Finnish map.

6.5 Regional Map

- Scatter map showing 33 Finnish supermarket locations across 6 chains (S-ryhma, K-ryhma, Lidl, Aldi, Tokmanni, M-ketju).
- Regional statistics table: store density, market share, and food-security indicators by region.
- After a Multi-Store simulation, the map shows a risk-level overlay: Critical / High / Moderate / Low - matched by store name.
- Click 'Re-run Multi-Store Simulation' to return to the network tool.

TIP Run Multi-Store Network first, then click 'View Results on Regional Map' to see which Finnish regions are most exposed in your scenario. The two sections are fully linked.

7. Card 5: Export & PDF Report

The Export card centralises all download options. Requires a completed Interactive Demo or Scientific Analysis run.

7.1 CSV Downloads

Daily Aggregates	Revenue, waste, lost sales, stock, panic - one row per day per scenario.
Per-Product Stock	Units on shelf per SKU per day for both scenarios.
SCM Log	Supply-chain events: orders placed, deliveries, stockout flags.
Food Waste Log	Units wasted per SKU per day with CO2-equivalent estimate.
Policy Run	Results of any policy interventions applied in Card 4.
Full Bundle	All of the above merged into a single multi-sheet CSV.

7.2 Generate PDF Report

Click 'Generate PDF Report' to create a branded 7-section PDF:

- 1. Executive Summary - revenue contraction %, waste increase %.
- 2. Simulation Parameters - full sidebar settings at the time of the run.
- 3. Revenue Analysis - cumulative and daily revenue charts.
- 4. Supply Chain Summary - stockout events and replenishment cycles.
- 5. Policy KPIs - if any policy was active during the run.
- 6. Stress Test Ranking - if a stress test was completed in this session.
- 7. Methodology Note - model architecture and data sources.

TIP The PDF report embeds Matplotlib charts directly - no internet connection required. Run Scientific Analysis first to include confidence bands in the charts.

8. SecureFood Scenario Simulator

The SecureFood Scenario Simulator is a dedicated tool for Horizon Europe SecureFood project partners (Grant No. 101136583). It models climate-disruption impacts on Finnish dairy supply chains and consumer food security.

8.1 Accessing the Simulator

Click 'Scenario Simulator' at the top right of any page (next to the app title). A separate 'Scenario Instructions' button downloads the full PDF walkthrough document.

8.2 Perspective Selection

Supply Chain Actor	Focus on revenue, stockout losses, shelf availability, and replenishment.
Policy Maker	Focus on food insecurity (FIES), budget exhaustion, and equity indicators.

8.3 Scenario Configuration

Disruption Type	Drought / Flood / Heat Wave / Cold Snap - affects domestic dairy yield.
Duration (days)	Length of the climate disruption event.
Severity (0-1)	Proportional reduction in domestic dairy supply.
Import Dependency	Share of dairy met by imports - determines exposure to disruption.
Crisis Start Day	When the disruption begins within the simulation window.

8.4 Output Charts (9 panels)

- Inflation-Volume Decomposition: constant-price vs nominal revenue.
- Stockout Revenue Loss: daily and cumulative lost sales from empty shelves.
- Shelf Stock by Category: baseline (dashed) vs crisis (solid) for each dairy sub-type.
- Consumer Panic & Stockpile Pressure: panic level and hoarding index over time.
- Basket Fulfilment Rate by Income Group: % of shopping basket successfully purchased.
- FIES Severe Food Insecurity: FAO scale index by income bracket.
- Budget Exhaustion & Gini Access Index: inequality in food access during the crisis.
- Domestic vs Import Sales Volume: food sovereignty and supply-chain resilience indicator.
- Policy Scenario Comparison: side-by-side effect of three intervention options.

TIP Each chart includes an expandable analysis box with quantified findings and citations from peer-reviewed literature. These are suitable for policy briefs and project reports.

9. Model Architecture & Technical Notes

9.1 Framework

- Built on Mesa (Python ABM framework) - each simulation step corresponds to one shopping day.
- Consumer agents: one ConsumerAgent created per shopper per day; removed at end of day.
- Product agents: persistent ProductAgent objects managing stock, orders, and waste.
- Scheduler: RandomActivation - agents act in a random order each step.

9.2 Consumer Decision Model

- Theory of Planned Behaviour (TPB): attitude, subjective norm, and perceived behavioural control (PBC) determine purchase intention.
- Prospect Theory loss aversion: consumers over-weight stockout risk, amplifying panic buying. Default $\lambda = 2.25$ (Kahneman & Tversky 1979).
- DCE-derived preference weights: willingness-to-pay for Finnish origin, organic, low-fat, and packaging attributes (N = 116, Horizon Europe SecureFood survey 2024-2025).
- Income brackets: Low (<1500 EUR/mo), Mid (1500-3500 EUR), High (>3500 EUR) - determine budget constraints and price sensitivity.

9.3 Supply Chain Model

- (s, Q) inventory policy: order Q units when stock falls below reorder point s.
- Lead time: orders arrive after a configurable number of days.
- Perishable waste: units exceeding shelf life are removed from stock and logged.
- Multi-store extension: panic contagion uses proximity-decayed transfer; hub stores buffer emergency redistribution to depleted stores.

9.4 Data Sources

DCE Survey	116 Finnish respondents; willingness-to-pay for food attributes (2024-2025).
Store locations	33 Finnish supermarkets georeferenced via OpenStreetMap (2025).
FIES	FAO Food Insecurity Experience Scale methodology (FAO 2016).
Emission factors	Product-level CO2e estimates from European Environment Agency (2023).
Prospect Theory	Kahneman & Tversky (1979); loss aversion $\lambda = 2.25$ (default).

10. Glossary

ABM	Agent-Based Model - simulation where individual agents follow rules and interact.
Archetype	Consumer segment: Bargain Hunter, Quality Seeker, Health Conscious, Convenience Shopper.
Crisis period	The day range during which supply disruption and/or inflation shocks are active.
DCE	Discrete Choice Experiment - survey method to elicit consumer preferences.
FIES	Food Insecurity Experience Scale (FAO) - 8-question ordinal severity scale.
Fulfilment rate	Share of a consumer's intended basket that was successfully purchased.
Gini access index	Inequality measure applied to food access; 0 = perfect equality, 1 = max inequality.
Hoarding multiplier	Factor scaling how much extra stock panic-buyers attempt to purchase.
Lead time	Days from replenishment order placement to shelf arrival.
Lost sales	Revenue foregone because a product was out of stock when a consumer wanted it.
Monte Carlo	Statistical method: run the simulation many times, report percentile bands.
Panic level	Agent-level indicator (0-1) reflecting perceived stockout risk and urgency to hoard.
PBC	Perceived Behavioural Control - TPB construct measuring ease of action.
Reorder point	Stock level (units) that triggers a new supply order.
SKU	Stock Keeping Unit - a unique product identifier in the catalogue.
TPB	Theory of Planned Behaviour: attitude + subjective norm + PBC gives intention.
Waste	Units removed from the model because they exceeded their shelf-life date.

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